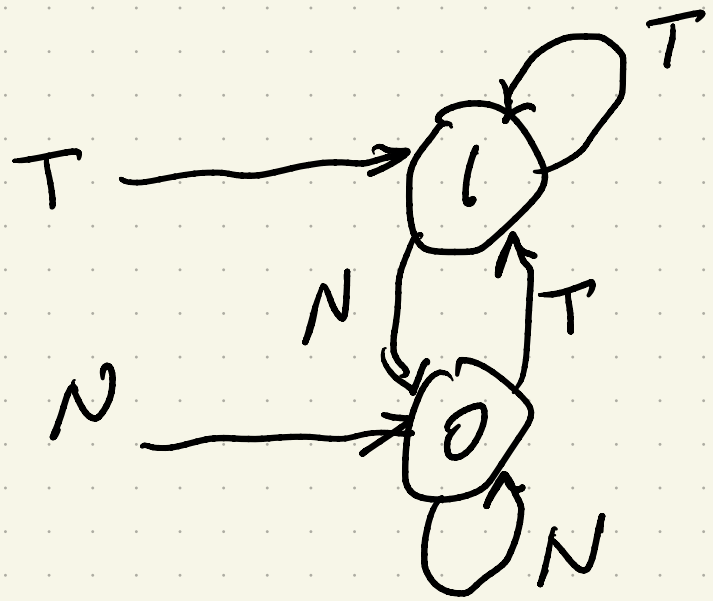
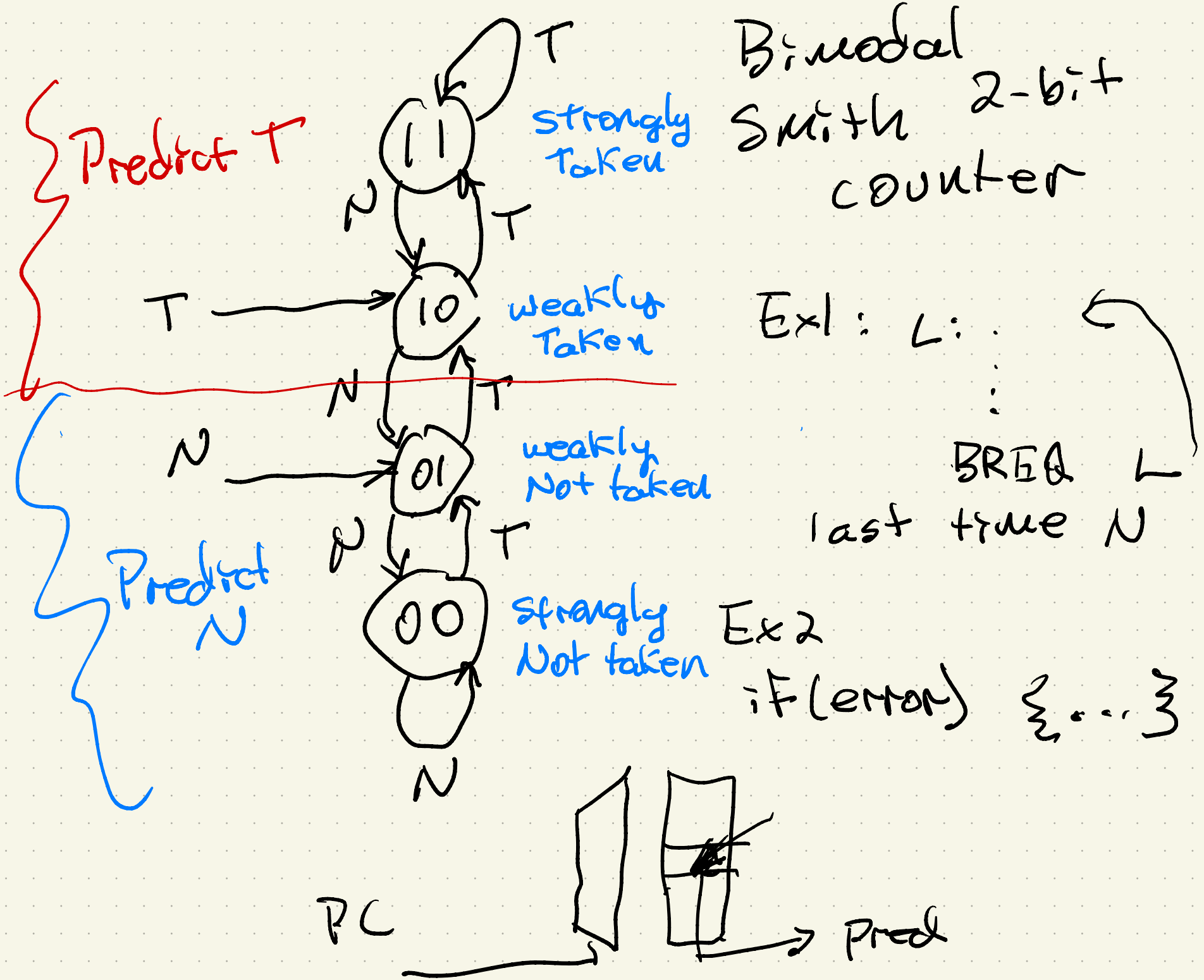
More sophisticated branch "whether" prediction



Our valid bit


Bimodal 2-bit Smith counter
Predict T
Predict N
11
strongly Taken
10
weakly Taken
01
weakly Not taken
00
Strongly Not taken
T
N
Ex1: L: ... BRNE L
last time N
Ex2
if(error) {...}
PC
Pred

Problem for Smith counter:

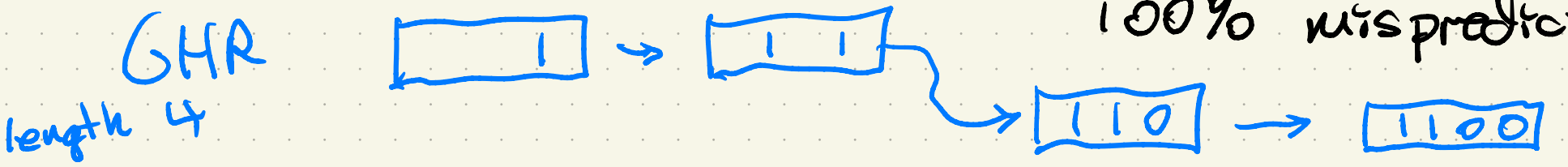
initially R1 = 0

```
A: BREQZ R1 D
D: BREQZ R1, F
F: NOT R1, R1
G: BR A
```

Smith at 01

A:	T	N	10
D:	T	N	10
A:	N	T	01
D:	N	T	01
	⋮		

100% misprediction



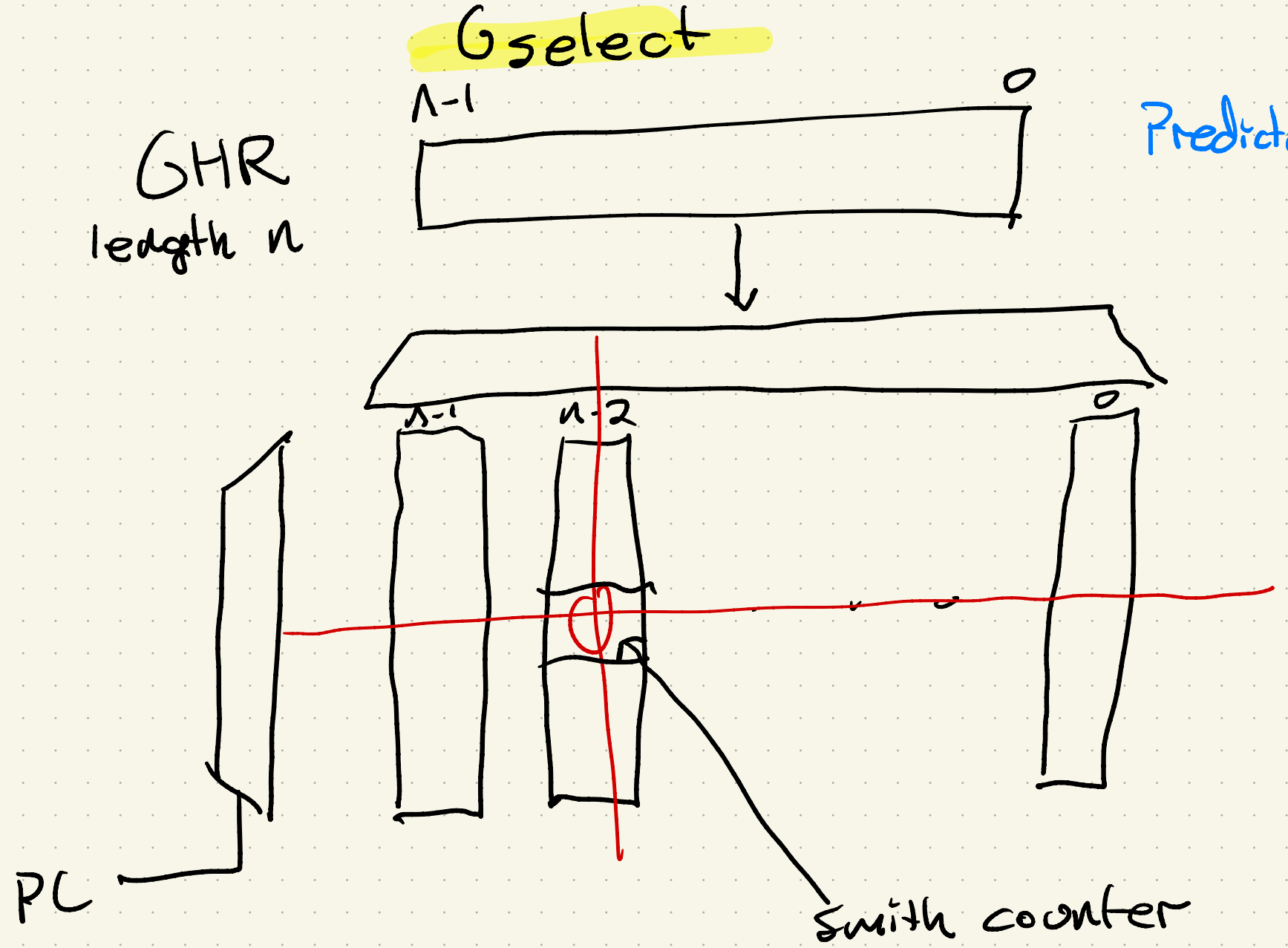
Capture correlation:

Global History Register

Shift register

if most recent known branch is Taken, shift in a 1, else shift in a 0

Gselect

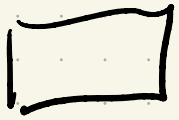


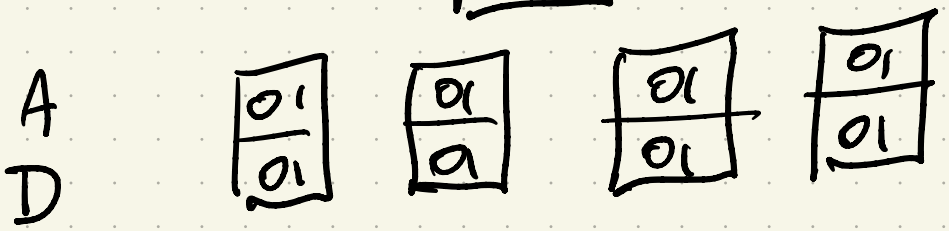
Predictor[GHR][PC]

initially R1 = 0

```
A: BREQZ R1 D
D: BREQZ R1, F
F: NOT R1, R1
G: BR A
```

GHR length 2 bits





GHR = 00

	0	1	2	3
A	01	01	01	01
D	01	01	01	01

A: pred N, Actually T

GHR 01

	0	1	2	3
A	10	01	01	01
D	01	01	01	01

D: pred N, T

GHR 11

	0	1	2	3
A	10	01	01	01
D	01	10	01	01

A: pred N, N

GHR 10

	0	1	2	3
A	10	01	01	00
D	01	10	01	01

D: pred N, N

GHR 00

	0	1	2	3
A	10	01	01	00
D	01	10	00	01

A: pred T, T

01

	0	1	2	3
A	11	01	01	00
D	01	10	00	01

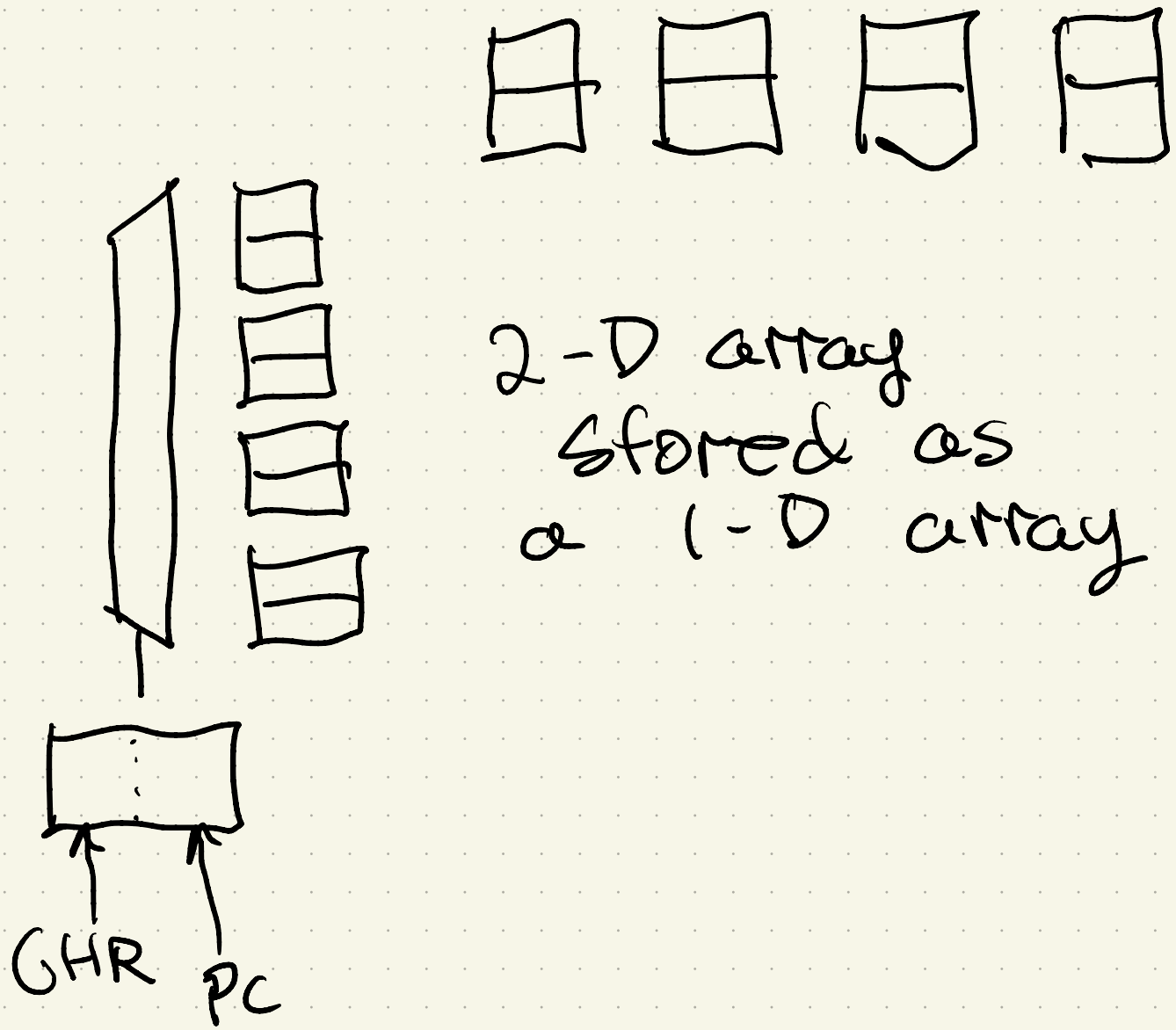
D: pred T, T

11

	0	1	2	3
A	11	01	01	00
D	01	11	00	01

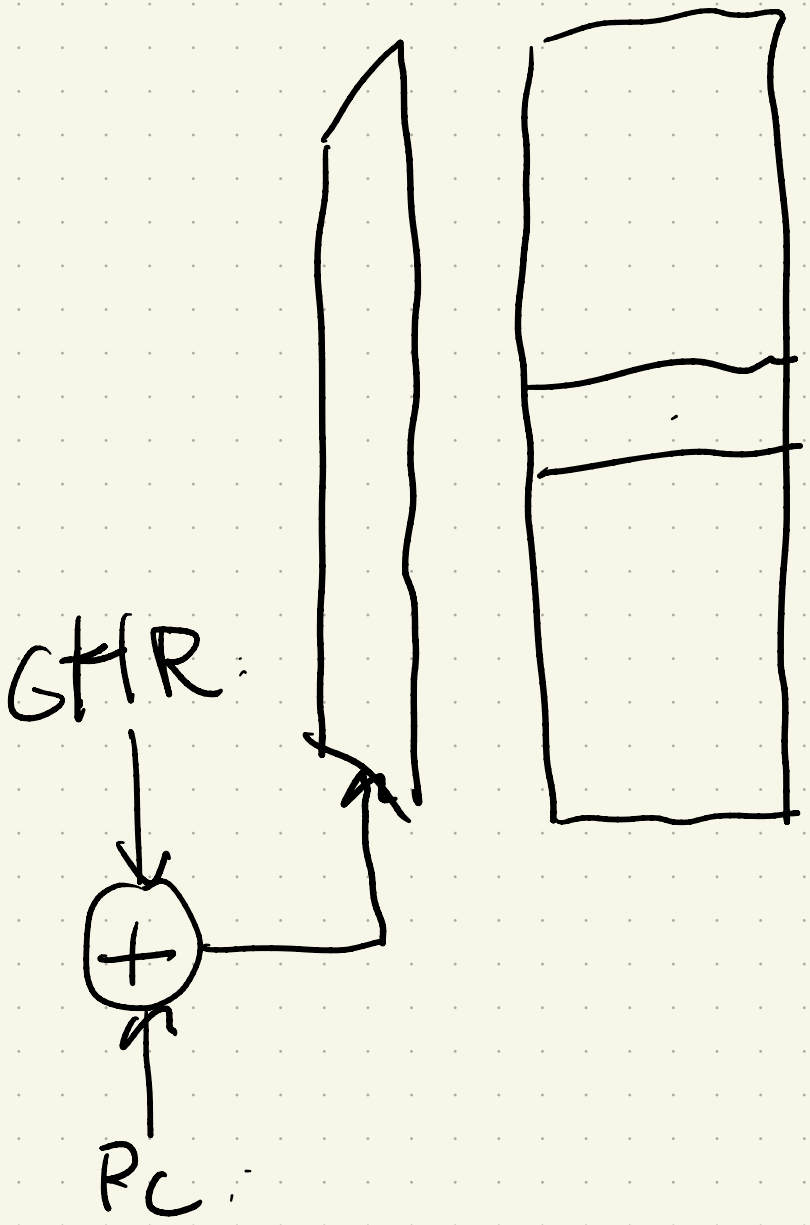
A: pred N, N

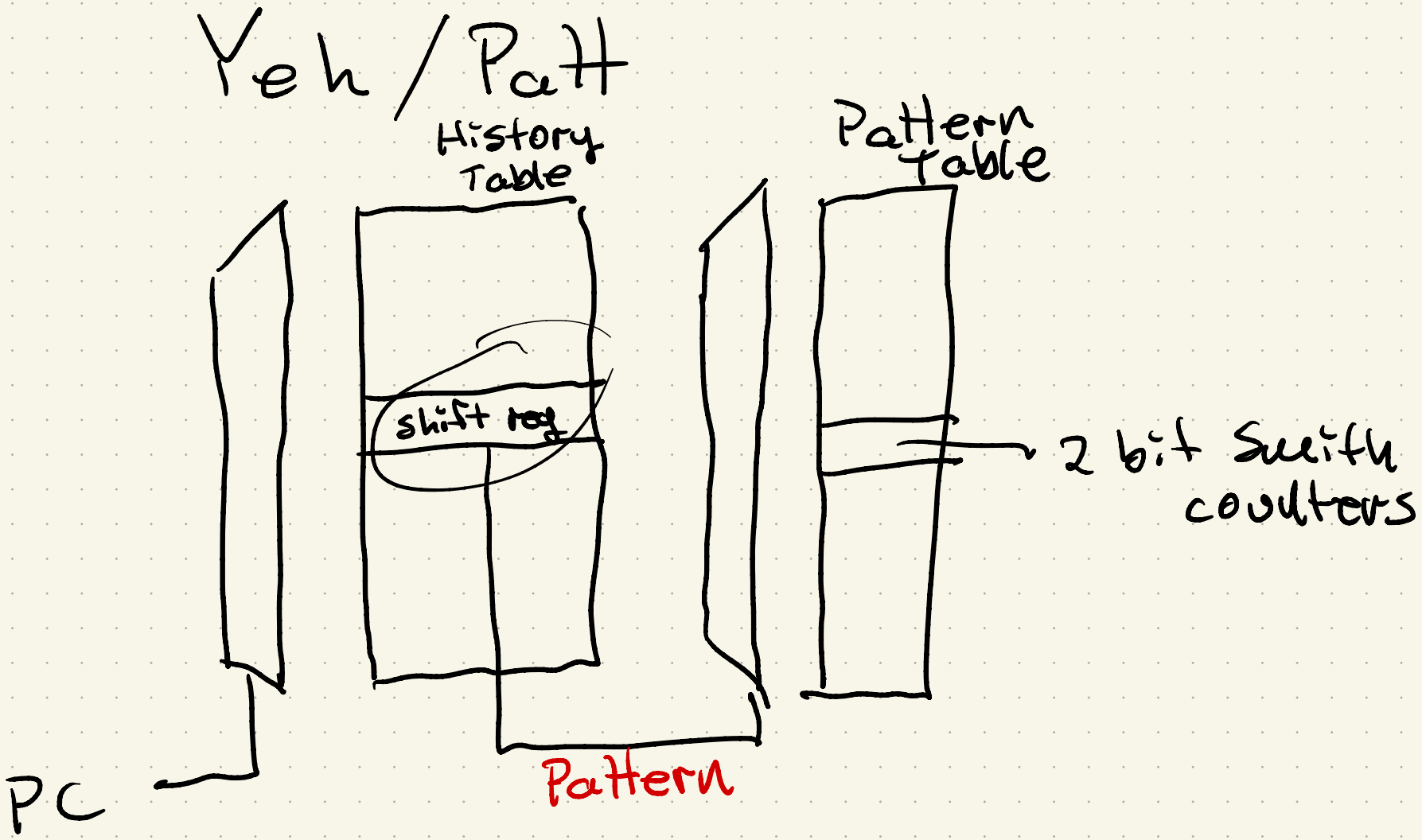
. . .



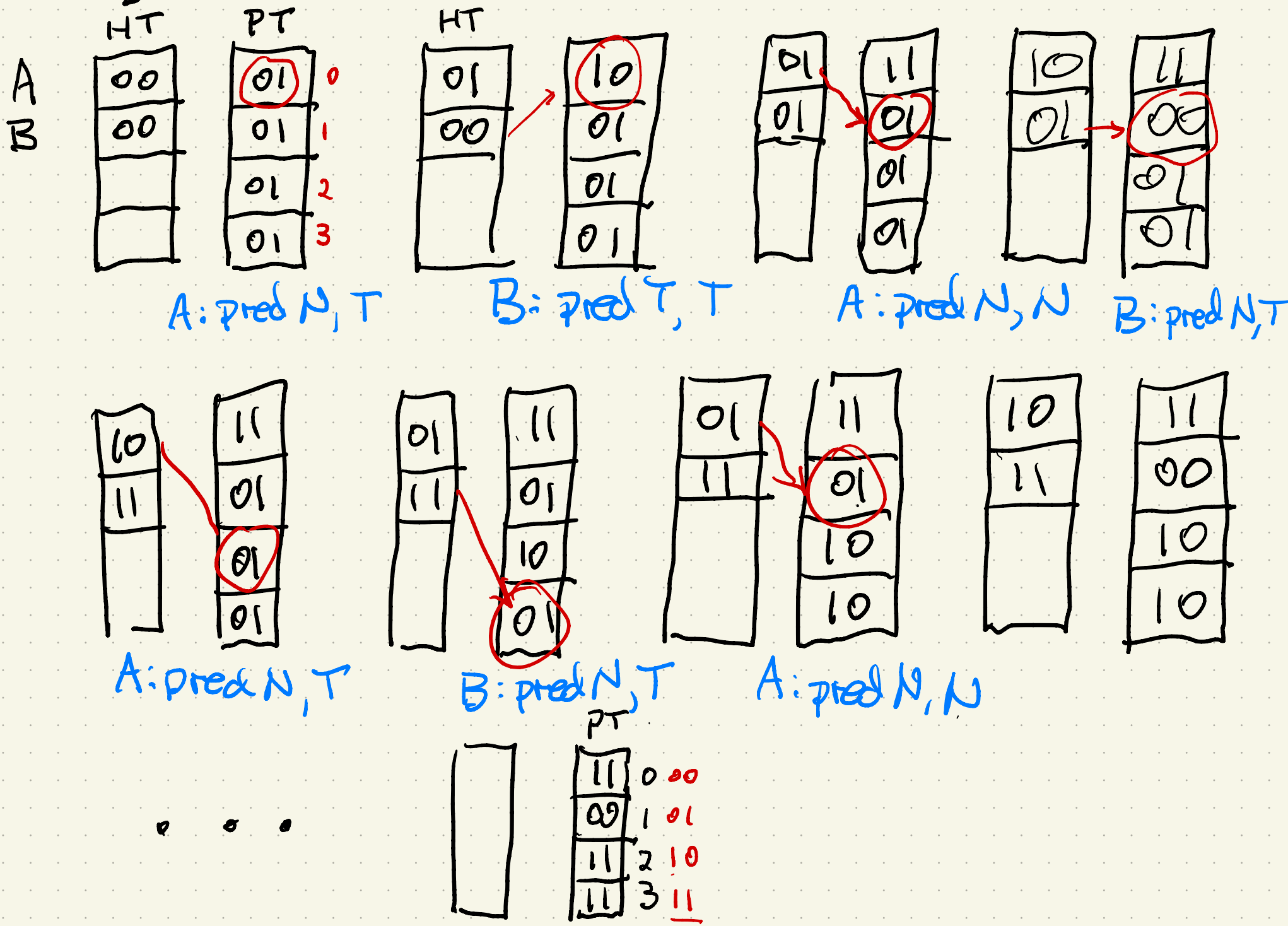
2-D array
stored as
a 1-D array

Gshare

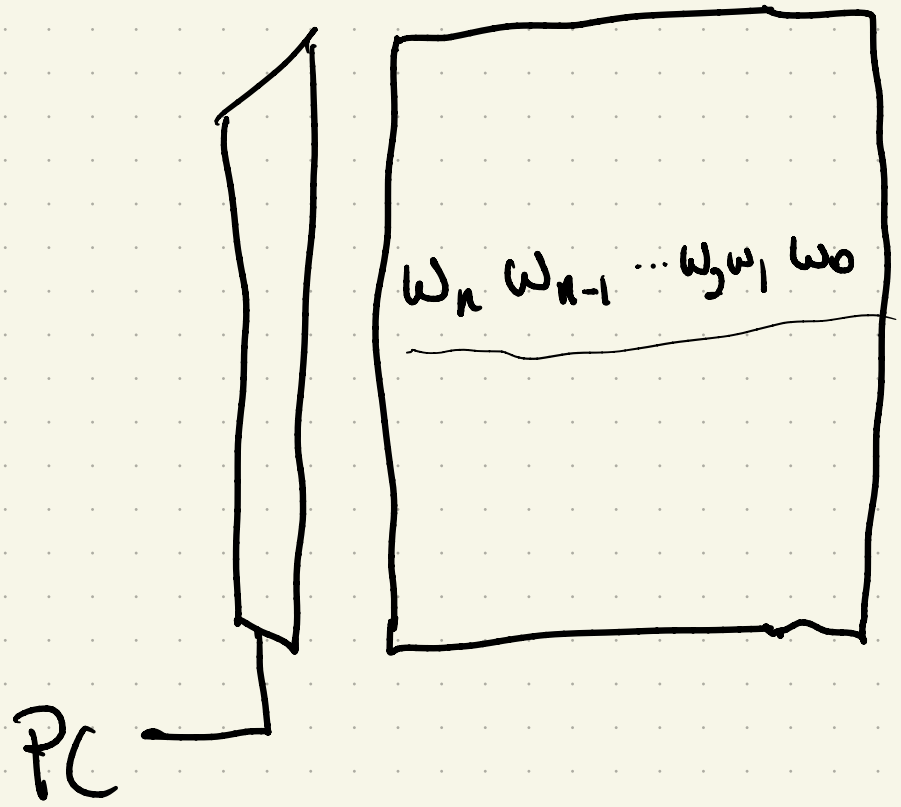



Yeh / Patt
History Table
Pattern Table
shift reg
2 bit Smith counters
PC
Pattern

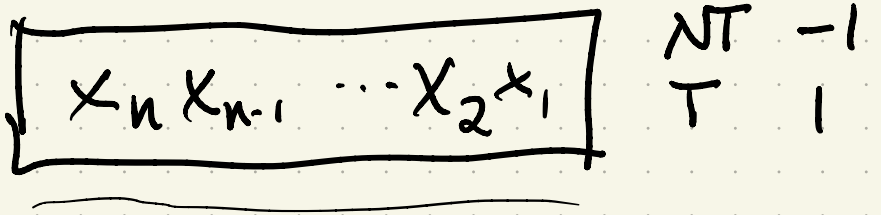
A: T N T N T N T N . . .
B: T T T T T T T . . .



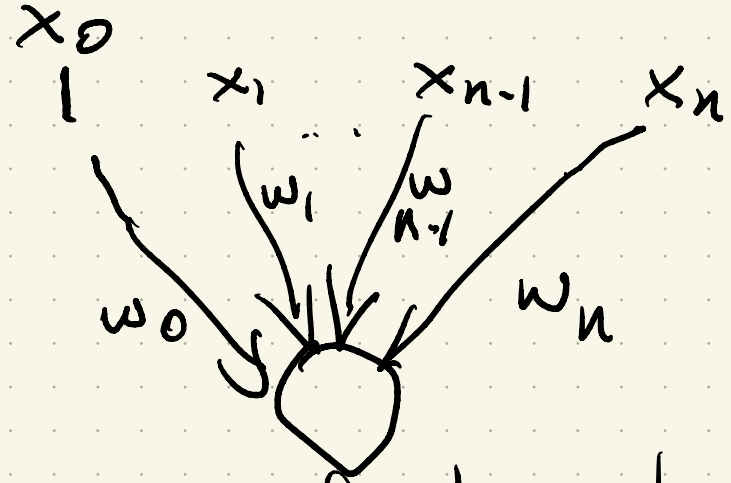
Perceptron Predictor



GHR



$$y = x_n w_n + x_{n-1} w_{n-1} + \cdots x_2 w_2 + x_1 w_1 + w_0$$



if $y > 0$ then predict taken
else not taken

if prediction != actual

or

$|y| < \text{theta}$ then

for i = 0 to n

$w_i = w_i + (\text{actual}) * x_i$

where $x_0 = 1$, actual is (NT: -1, T: 1)

Ex) GHR -1, 1, 1, -1

Perceptron 1, 1, 1, 1, 1

pred = -1*1 + 1*1 + 1*1 + 1*1 - 1*1 = 1 > 0 pred Token

Actual is NT (-1) 2, 0, 0, 2, 0

New perceptron $w0 = 1 + (-1) \cdot 1 = 0$

$w1 = 1 + (-1) \cdot -1 = 0$